Computer Vision: Teaching Machines to See

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Introduction to Computer Vision

Computer vision is a field of artificial intelligence that enables computers to understand and interpret images and videos. Humans see effortlessly, but for a computer an image is only a grid of numbers. A colour photograph is stored as pixels, and each pixel has three values for red, green and blue, usually between 0 and 255. Computer vision turns these numbers into meaningful information such as "this is a face", "there are three cars" or "this tyre has a crack". Computer vision matters because cameras are everywhere, in phones, vehicles, factories, hospitals and satellites, and automatic understanding of what they capture saves time, improves safety and opens new applications.

Brief History

Computer vision research began in the 1960s. In 1966 an MIT summer project aimed to connect a camera to a computer and describe what it saw, which proved far harder than expected. Yann LeCun developed LeNet, an early convolutional neural network for reading handwritten digits, in the late 1980s and 1990s. In 2001 the Viola-Jones method made real-time face detection practical. The ImageNet dataset, led by Fei-Fei Li and released in 2009, gave researchers over a million labelled images. In 2012 the AlexNet convolutional neural network won the ImageNet challenge by a large margin and started the deep learning revolution in vision. Later models include VGG, ResNet, YOLO, Faster R-CNN and, from 2020, Vision Transformers.

Core Tasks: Image Classification

Image classification is the task of assigning a single label to a whole image. The input is an image and the output is a category such as "cat", "dog" or "car", usually with a confidence score. For example, a model may look at a photo and say it is 97 percent likely to show a dog. Classification tells us what is in the picture but not where it is. It is used for sorting photos, identifying plant diseases from a leaf picture, and screening medical images such as chest X-rays as normal or abnormal.

Object Detection and How It Differs from Classification

Object detection goes a step further than classification. It finds multiple objects in an image, draws a bounding box around each object to show its location, and labels each box with a class. So classification answers "what is in this image?", while object detection answers "what objects are in this image and where are they?" An image of a street might contain several cars, pedestrians and traffic lights, and an object detector marks every one of them separately. Popular detectors are YOLO (You Only Look Once), SSD and Faster R-CNN. Detection quality is measured with Intersection over Union (IoU), which compares the predicted box with the true box, and with mean Average Precision (mAP). Small-object detection, for example detecting distant vehicles in drone images, is an especially hard research problem.

Other Computer Vision Tasks

Semantic segmentation labels every pixel in an image by class, such as road, sky or person. Instance segmentation separates individual objects of the same class. Object tracking follows objects across video frames. Pose estimation finds body joints. Optical Character Recognition (OCR) reads printed or handwritten text, and face recognition identifies people. Image generation and image captioning combine vision with language.

How Computer Vision Works

Most modern systems use Convolutional Neural Networks (CNNs). A CNN slides small filters over an image to detect edges, then textures, then shapes and finally whole objects. Layers called pooling layers shrink the data, and final layers produce the answer. To train a model, thousands of labelled images are needed; data augmentation, such as flipping, rotating and changing brightness, creates more variety. Transfer learning reuses a model already trained on a large dataset and adapts it to a new task, which saves data and time.

Computer Vision in Cars

Two uses of computer vision in cars are lane detection and pedestrian or obstacle detection. Cameras read lane markings so the car can warn the driver or steer to stay in lane, and detectors identify pedestrians, cyclists and other vehicles to trigger automatic emergency braking. Other uses include traffic sign recognition, driver drowsiness monitoring through eye tracking, parking assistance with surround-view cameras, and adaptive cruise control. Fully self-driving systems combine cameras with radar and LiDAR. Tesla Autopilot and Waymo are well-known examples.

Uses and Indian Examples

Computer vision is used in medical imaging, factory quality inspection, retail checkout, agriculture and security. In India, Automatic Number Plate Recognition on highways and city traffic cameras, face authentication for Aadhaar, FASTag and toll monitoring, drone-based crop monitoring, and AI screening of diabetic eye disease in hospitals are real examples. Chennai companies and research groups work on industrial defect detection and small-object detection in aerial images.

Challenges and Future

Challenges include poor lighting, occlusion, small or blurred objects, bias in face recognition, privacy concerns and the need for labelled data. The future includes vision-language models that understand images and text together, efficient models running on edge devices, and 3D vision.

Quick Facts

- Image classification assigns one label to a whole image.
- Object detection locates multiple objects with bounding boxes and labels each one.
- Two uses of computer vision in cars are lane detection and pedestrian detection.
- AlexNet won the ImageNet challenge in 2012 and started the deep learning era in vision.
- Computers see images as grids of pixel numbers, each pixel typically having red, green and blue values from 0 to 255.